

Discipline is more important than motivation

12-Month Roadmap to Become an Expert in Agentic AI for Finance & Investment Banking

This roadmap is designed specifically for:

- Your finance + regulatory background
- Your existing exposure to data lakes / analytics
- Your goal of becoming strong in Agentic AI systems
- Your current starting point (no NLP/LLM knowledge yet)

The roadmap is intentionally:

- Deep
- Practical
- Architecture-focused
- Project-driven
- Industry-aligned

You will gradually evolve from:

beginner → AI engineer → RAG engineer → agentic systems engineer → finance AI architect

OVERALL TARGET AFTER 12 MONTHS

By the end, you should be able to build:

- ✓ Enterprise RAG systems
 - ✓ Multi-agent AI workflows
 - ✓ Financial AI copilots
 - ✓ Regulatory AI systems
 - ✓ Autonomous document analysis pipelines
 - ✓ Production-grade AI applications
 - ✓ AI architectures for banking/finance
-

MONTH 1 — Python + APIs + Developer Foundations

Goal

Become comfortable writing code daily.

Learn

Python

Topics:

- variables
 - loops
 - functions
 - classes
 - OOP
 - dictionaries
 - JSON
 - exception handling
 - file handling
-

Tools

- VS Code
 - Git
 - GitHub
 - virtual environments
 - pip
-

APIs

Learn:

- REST
- GET/POST

- headers
 - authentication
 - JSON parsing
-

Build

Project 1 — Financial Data Fetcher

Use APIs to fetch:

- stock data
- SEC filings
- market news

Store results in CSV/JSON.

Project 2 — Earnings News Analyzer

Input:

- financial news

Output:

- summarized insights

(No LLM yet — use Python processing.)

Resources

Python

- [Python Official Tutorial](#)

Git

- [GitHub Skills](#)
-

MONTH 2 — AI/LLM Foundations

Goal

Understand modern LLM systems.

Learn

Concepts

- NLP basics
- tokens
- embeddings
- transformers
- attention
- prompts
- temperature
- hallucinations
- context windows

APIs

Learn:

- [OpenAI Platform](#)
- [Anthropic Docs](#)

Build

Project 1 — Financial Chatbot

Capabilities:

- answer finance questions
- summarize earnings reports

Project 2 — SEC Filing Summarizer

Input:

- 10-K PDF

Output:

- risks
 - opportunities
 - management commentary
-

Important Skill

Learn prompt engineering deeply:

- system prompts
 - structured outputs
 - JSON mode
 - function calling
-

MONTH 3 — Embeddings + Vector Search

Goal

Understand semantic search deeply.

This is foundational for RAG.

Learn

Concepts

- embeddings
 - cosine similarity
 - vector search
 - ANN indexing
 - metadata filtering
-

Vector Databases

Learn:

- [Chroma](#)
- [Pinecone](#)

Build

Project — Financial Semantic Search Engine

Upload:

- annual reports
- SEC filings
- research reports

Features:

- semantic search
 - source references
 - filtering by company/year
-

Deep Focus

Understand:

- why embeddings work
 - chunking tradeoffs
 - token limits
 - retrieval quality
-

MONTH 4 — RAG Systems (Core Month)

Goal

Become strong in Retrieval-Augmented Generation.

This is one of the most important months.

Learn

RAG Architecture

- ingestion
- chunking

- retrieval
 - reranking
 - prompt assembly
 - citations
-

Frameworks

Learn:

- [LangChain](#)
 - [LlamaIndex](#)
-

Build

Project — Investment Banking Knowledge Assistant

Capabilities:

- ask questions across documents
- compare companies
- summarize filings
- cite sources

Documents:

- annual reports
 - M&A documents
 - earnings transcripts
-

Advanced Topics

Learn:

- hybrid search
- recursive chunking
- metadata filtering
- query rewriting

MONTH 5 — Advanced RAG + Evaluation

Goal

Move from beginner RAG to enterprise RAG.

Learn

Advanced RAG

- parent-child retrieval
 - graph RAG
 - rerankers
 - multi-hop retrieval
 - query transformation
 - agentic RAG
-

Evaluation

Learn:

- retrieval precision
 - hallucination detection
 - groundedness
 - answer faithfulness
-

Build

Project — Regulatory Intelligence Platform

Use your domain expertise.

Features:

- retrieve regulations
- compare rules
- answer compliance questions

- explain reporting logic
-

Important

Start writing:

- architecture diagrams
- technical READMEs
- system flow documentation

This matters professionally.

MONTH 6 — Agentic AI Foundations

Goal

Understand how AI agents work.

This is where your specialization begins.

Learn

Concepts

- planning
 - memory
 - tool use
 - reflection
 - task decomposition
 - agent loops
 - reasoning chains
-

Frameworks

Learn:

- [CrewAI](#)
- [Microsoft AutoGen](#)

Build

Project — Multi-Agent Financial Research Team

Agents:

- market agent
- SEC agent
- valuation agent
- news agent
- report writer

Key Learning

Understand:

- coordination
- orchestration
- delegation
- memory handling

MONTH 7 — LangGraph + Workflow Systems

Goal

Learn enterprise orchestration.

Learn

LangGraph

- state machines
- workflows
- branching
- memory
- retries

- checkpoints

Learn:

- [LangGraph](#)
-

Build

Project — Autonomous Reporting Workflow

Workflow:

1. ingest data
2. validate
3. detect anomalies
4. retrieve regulations
5. generate narrative
6. send approval

This directly aligns with enterprise finance systems.

Important Skill

Learn:

- human-in-the-loop systems
 - approval workflows
 - auditability
-

MONTH 8 — Backend + Production Engineering

Goal

Become capable of building deployable AI systems.

Learn

Backend

- FastAPI

- authentication
- async APIs
- websockets

Learn:

- [FastAPI](#)
-

Databases

- PostgreSQL
 - Redis
-

Docker

Learn:

- containers
 - Dockerfiles
 - deployment basics
-

Build

Project — AI Financial API Platform

Expose:

- RAG endpoints
 - agent endpoints
 - workflow endpoints
-

Important

Learn:

- API architecture
- rate limiting
- caching

- logging
-

MONTH 9 — Cloud + Enterprise AI

Goal

Learn enterprise-grade deployment.

Learn

Azure AI Ecosystem

Especially relevant for your background.

Learn:

- [Azure OpenAI Service](#)
 - [Microsoft Azure AI Foundry](#)
-

Learn

- model deployment
 - vector indexing
 - AI governance
 - RBAC
 - secrets management
-

Build

Project — Enterprise Financial Copilot

Capabilities:

- secure authentication
 - role-based access
 - financial knowledge retrieval
 - workflow automation
-

MONTH 10 — AI Observability + Governance

Goal

Learn production AI reliability.

This is VERY important in finance.

Learn

Topics

- tracing
 - prompt logging
 - evaluation pipelines
 - hallucination detection
 - AI governance
 - guardrails
 - prompt injection defense
-

Tools

Learn:

- [LangSmith](#)
-

Build

Project — Auditable AI Reporting System

Features:

- citations
 - traceability
 - approval logs
 - evaluation scoring
-

MONTH 11 — Advanced Agentic Architectures

Goal

Become advanced.

Learn

Topics

- MCP (Model Context Protocol)
 - memory architectures
 - graph databases
 - event-driven agents
 - long-running agents
 - autonomous workflows
-

Explore

- [Model Context Protocol](#)
-

Build

Project — Autonomous Investment Research Platform

Features:

- multi-agent planning
 - long memory
 - research automation
 - portfolio analysis
 - market monitoring
-

MONTH 12 — Capstone + Professional Positioning

Goal

Create a world-class portfolio.

Build Your Capstone

Ultimate Project

Agentic AI Platform for Financial Intelligence

Features:

- RAG
 - multi-agent orchestration
 - regulatory intelligence
 - portfolio analysis
 - reporting workflows
 - governance
 - audit trails
 - dashboards
 - approvals
-

Create

GitHub Portfolio

Include:

- architecture diagrams
 - setup instructions
 - demos
 - screenshots
 - videos
-

Write Technical Articles

Topics:

- AI in regulatory reporting
- Agentic workflows in banking
- Financial RAG systems

- Enterprise AI governance
-

Build Your Professional Identity

Position yourself as:

“Agentic AI Engineer specializing in Finance & Investment Banking Systems”

That niche is extremely valuable.

DAILY STUDY STRUCTURE

Weekdays (2–3 hrs/day)

1 hr

Coding

45 mins

Reading/documentation

30 mins

Architecture understanding

30 mins

Project work

Weekends (5–8 hrs)

Focus on:

- building
 - debugging
 - deployment
 - project completion
-

IMPORTANT BOOKS

Python

- Fluent Python

LLMs

- Hands-On Large Language Models

System Design

- Designing Data-Intensive Applications
-

MOST IMPORTANT CAREER ADVICE

The winners in Agentic AI will NOT just be:

- prompt engineers
- chatbot builders

The winners will be people who can combine:

- domain expertise
- workflows
- enterprise systems
- governance
- AI orchestration

Your finance + regulatory background is a major strategic advantage.

What You Should Avoid

Avoid spending months on:

- pure ML math
- low-level transformer research
- academic NLP papers initially

You can learn those later if needed.

Your Strongest Future Niche

You are uniquely positioned for:

- AI for regulatory reporting
- AI for treasury

- AI for compliance
- AI for financial operations
- AI workflow automation in banking

That is an extremely strong market position.